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Operando High-Resolution Swell Mapping for Mixed Ionic-Electronic Channel

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Abstract

Organic electrochemical transistors (OECTs) and related bioelectronics operate through ionic (de)doping of organic mixed ionic-electronic (semi)conductor (OMIEC) channels with concomitant swelling. However, operando swelling monitoring with high spatiotemporal resolution has remained a challenge owing to the inherent limitations of existing techniques. Here, we introduce an in-situ characterization platform for real-time swelling monitoring of OECT channels by incorporating a laser Doppler vibrometer with a customized testing module. This approach enables dynamic reconstruction of the swelling process with sub-nanometer and sub-microsecond spatiotemporal resolution, revealing apparent inhomogeneity in swelling magnitudes (4 nm ~ 400 nm) in different OMIECs and channel locations, along with distinct ionic (de)doping paths originating from vertical channel edge. Moreover, we identify the Coulomb force between vertically stacked source/drain electrodes as an effective factor for swelling suppression, which enables the demonstration of high-fidelity OECTs (> 15 million full switching cycles) and organic artificial neurons (> 15-day stable operation in 1×PBS). Our work provides a brand-new route for guiding OMIEC synthesis, elucidating underlying mechanisms, and advancing the design of robust (bio)electronic devices.

Main

Organic electrochemical transistors (OECTs), which show tremendous potential in neuromorphic electronics[1-3], brain-machine interfaces[4,5], and implantable sensors[6,7], depend mainly on organic mixed ionic-electronic conductor (OMIEC) channels. While ionic-electronic coupling of OMIECs leads to apparent swelling, which is essential for device function[8-10]. Therefore, swelling characteristics of OMIEC channels are extremely important for ionic-electronic coupling exploration and material/device performance optimization[11-14]. Various techniques, including electrochemical quartz crystal microbalance with dissipation monitoring (EQCM-D)[15], electrochemical strain microscopy (ESM)[16], four-dimensional scanning transmission electron microscopy (4D-STEM)[17], atomic force microscopy (AFM)[18], and grazing-incidence wide-angle X-ray scattering (GIWAXS)[19], have been developed to study OMIEC swelling. Nevertheless, GIWAXS, ESM, 4D-STEM, and AFM enable atomic-level microstructure characterization, but are not compatible with dynamic measurements. While EQCM-D holds advantages in dynamically estimating OMIEC mass variations, it is limited in temporal and spatial resolutions. Therefore, in-situ swelling monitoring on operating OECTs with desired temporal (down to milliseconds) and spatial (down to micrometers) resolutions is not accessible, which severely hinders a comprehensive understanding of OMIEC swelling and restricts further material/device optimization and related bioelectronic development.

Here, we demonstrate an in-situ channel swelling monitoring platform by constructing a laser Doppler vibrometer (LDV) with a customized monitoring platform holding a vertical OECT (vOECT) structure with a side-gate configuration (named as laser Doppler vibrometer for swell mapping, LDV-M), enabling high temporal (<0.1 ms) and spatial (x-y axis ~3 μm, z axis ~0.6 nm) resolution, along with a wide measurable

swelling range up to 1 mm. Moreover, LDV-M is the only platform that can enable in-situ dynamic swelling monitoring (Extended Data Table 1), which is further applied for high-resolution swelling mapping, enabling direct observation of swelling heterogeneity across different OMIECs under different operation conditions (switching frequencies, driving voltages, cycling status, etc). Notably, we unveil previously inaccessible swelling pathways and transient behavior, identifying apparent edge-dominated swelling (up to 389%) that can be effectively inhibited via the Coulomb force between the vertically stacked source and drain electrodes. Leveraging this discovery, by optimizing device encapsulation, enhanced vOECT cyclic stability for more than 15 million full cycles is demonstrated. Furthermore, vOECT-based artificial neurons are fabricated, showing record-high firing frequency (covering the firing spectrum of all human neurons) and sustainability among reported OECT-based organic neurons. These findings provide pivotal guidance for high-fidelity bioelectronic material and device engineering.

Design and configuration of the swelling dynamic monitoring system

The LDV-M primarily comprises a laser Doppler vibrometer, a magnification objective lens, and a planar electrical positioning stage (Fig. 1a). Leveraging the laser Doppler shift effect and interference principle[20,21], this system enables non-contact dynamic swelling measurement of the vOECT channel. Specifically, the laser emitted by the laser source is split into reference light and measurement light by a beam splitter. The measurement light, launched through the objective lens, focuses onto the electrode surface of the device to form a 3 μm-diameter spot. Concurrently, a gate voltage (V_G) is applied to the device to induce doping/dedoping processes. Thus, the resulting channel swelling alters the phase of the reflected light wavefront, generating a Doppler shift. Subsequently, the reflected light returns to the laser source and interferes with the reference light (with a fixed optical length)[22]. The interference signal captured by the detector contains the device's swelling displacement information, which is then decoded via digital demodulation techniques to output the swelling displacement magnitude (with a resolution of ~0.6 nm, limited mainly by the surrounding environmental vibration noises). Notably, by coordinating with the planar electrical positioning stage (~ 1 μm x/y-axis movement resolution), swelling signals at different positions on the device can be measured, enabling dynamic channel swell mapping. This LDV-M meets the testing requirements for OECTs with micrometer-scale channel dimensions and sub-millisecond-level response times (Fig. 1b). Photographs of the overall LDV-M configuration, along with the zoomed-in vOECT mounting module and vOECT channel with the laser spot focused on, are presented in Fig. 1c-e. The composition of each subsystem is comprehensively detailed in Methods.

Single-point swelling monitoring of vOECTs with different OMIECs

A side-gated vOECT architecture with 50 μm top (drain) and bottom (source) electrode width is employed (Supplementary Figs. 1 and 2)[23]. A detailed fabrication process can be found in Methods. For the OMIECs, two n-type (BBL, Homo-gDPP) and two p-type (p(g2T-T), gDPP-g2T) materials are adopted, where Homo-gDPP, p(g2T-T), and gDPP-g2T, respectively, are blended with a crosslinker [DtFDA (DA) or DtFGDA (GDA)] holding a mass ratio of 5:1 (Supplementary Figs. 3) [24-28]. First, the transistor

performances of all vOECTs are evaluated (Fig. 2a,b, and Extended Data Fig.1), where ~10^5 or higher current on/off ratios and mA-level on currents (I_{on}) are obtained in transistors with different channels.

Next, single-point swelling monitoring on these vOECTs is conducted with the laser point located approximately at the channel center (Supplementary Figs. 4), where detailed experimental and data analysis methods can be found in Methods. As shown in Fig. 2c,d, and Extended Data Fig.2a-f, the LDV-M demonstrates high temporal resolution and precise z-axis swelling monitoring capability under three different V_G switching frequencies (3.3, 11.1, and 111.1 Hz). In the BBL-based channel (Fig. 2c), under a 3.3 Hz switching frequency, active swelling progressively increases during doping along with rising drain current (I_D), reaching maximum active swelling concurrent with I_D saturation, followed by contraction during dedoping (falling I_D). Increasing the frequency to 11.1 Hz yields a comparable active swelling amplitude to that under 3.3 Hz, attributed to complete doping/dedoping processes under both frequencies. Further increase to 111.1 Hz results in incomplete doping/dedoping as demonstrated by I_D versus time, accompanied by significantly reduced active swelling. For the p-type p(g2T-T) channel, the active swelling behavior also exhibits a strong correlation with the degree of doping/dedoping (the amplitude of I_D) as the device switching frequency varies (Fig. 2d).

All channels with varying materials demonstrate similar active swelling trends, but with different characteristics (Extended Data Fig.2a-f). First, active swelling magnitudes vary tremendously in channels with different materials (Fig. 2e), where n-type vOECTs show overall stronger active swelling. For instance, under 3.3 Hz, 115.9±1.5 nm active swelling is observed in BBL-based vOECT, while only 4.1±0.2 nm is observed in p(g2T-T): DA-based one. This is further verified by EQCM (Supplementary Figs. 5), which shows that the BBL film exhibits a highly pronounced active swelling compared to p(g2T-T), attributable to the ladder polymer structure requiring substantial water influx during doping[18]. Second, when replacing DA with more hydrophilic GDA crosslinker, pronounced active swellings are observed along with faster switching characteristics in p(g2T-T), Homo-gDPP, and gDPP-g2T based vOECTs (Fig. 2e and Supplementary Table 1). Third, when cycling frequency increases, not only is a reduced active swelling amplitude observed, but an overall swelling baseline shift to a higher z plateau is revealed (Supplementary Figs. 6). This indicates that when operating under high switching frequencies, the injected electrolyte cannot be fully ejected from the channel (*vide infra* for swell mapping).

Next, the transient swelling behavior is analyzed in detail. In the BBL-based vOECT (Fig. 2f), swelling waveforms are consistent with I_D curves across all frequencies, transitioning from a near square-wave shape at low frequencies to a triangular-wave shape at high frequencies. Furthermore, the active swelling rate during the doping process is overall slower than that during the dedoping process, which correlates directly with the I_D behavior and agrees well with the reported “slow on, fast off” behavior (Supplementary Figs. 7)[29]. Notably, when switching from off to on or from on to off, a sharp decrease or increase spike in I_D is observed due to the charging or discharging of the capacitance[24]. While at the same moment, the film active swelling exhibits a “hold” pattern, showing no swelling. However, this is not attributed to capacitance charging/discharging, but due to delayed active swelling of the channel

center (*vide infra* for swell mapping). Similar active swelling characteristics are observed in p(g2T-T): DA-based vOECT, with one key difference: after the “hold” pattern during the off-to-on process, the film undergoes transient contraction before expansion (Fig. 2g). As shown in Supplementary Figs. 5, unlike that in BBL, p(g2T-T) reveals obvious active swelling either biased positively or negatively, indicating good ionic conductivity for both cations and anions[30]. Consequently, when switching from off to on, counterions [cations for p(g2T-T)] would be ejected first, leading to film contraction first. Then, anions would be injected, leading to film expansion. Such a phenomenon has also been observed in the gDPP-g2T-based channel (Extended Data Fig.2e,f). Note that such a system can also be applied for the swelling monitoring of planar OECTs, but requires good laser light reflection of the channel surface (Extended Data Fig.2g-i).

vOECT channel swell mapping

By precisely controlling the relative position between the channel and the laser spot using the planar electrical positioning stage (Fig. 3a), high temporal resolution of the single-point channel swelling monitoring can be incorporated with swell mapping capability across the entire vOECT channel. Specifically, a 6×4 measurement grid [(x, y), where x = 1:6, y = 1:4] is constructed over the channel region (80×50 μm^2) by scanning the laser spot in defined steps (Extended Data Fig.3). Note, points at (1, y) and (6, y) are located outside the channel with no bottom electrode underneath. Detailed testing protocols can be found in Methods.

Swell mappings of different channels are then conducted (Supplementary Videos 1 to 12), demonstrating the unique operando swelling monitoring capability of LDV-M. By extracting specific frames from the mapping movies, the channel swelling behavior can be analyzed with high spatiotemporal resolution. As shown in Fig. 3b,c, and Extended Data Fig.4a,b, obvious swelling in the channel area is observed upon switching, while apparent differences are observed in different channel locations and different driving frequencies. For instance, with 11.1 Hz switching frequency (Fig. 3c), at the initial doping stage (t_1 to t_2), swelling is more pronounced at the channel periphery than at the center; with time progression (t_3 to t_5), swelling propagates from channel edge to channel center, reaching saturation. Upon the dedoping (t_5 to t_8) process, the channel edges obviously contract faster than the channel central region. Such a spatiotemporal dependent swelling characteristic provides the hard evidence that, in vOECTs, the doping/dedoping process always starts from the channel edge and then evolves to the channel bulk. With a higher switching frequency of 111.1 Hz, the swelling difference between channel edge and channel center magnifies, indicating unsaturated doping of the channel bulk, which is further supported by the unsaturated I_D (Extended Data Fig.4b). Moreover, upon repeated switching cycles, a gradual swelling baseline shift towards a higher z plateau is also observed under 111.1 Hz switching frequency (Extended Data Fig.4c). Such a phenomenon suggests that both doping and dedoping processes cannot reach saturation under high switching frequencies due to inferior ionic mobility.

Next, active swelling behaviors at 3 representative positions are analyzed in detail, which are (1,3), (3,3), and (3,4), representing locations at top electrode edge with no bottom electrode underneath, top electrode edge with bottom electrode underneath, and top electrode above channel center, respectively (Fig. 3d). Obviously, active swelling transient behavior (under 11.1 Hz) at (3,3) is not only lagged (1,3) and (3,4), but also show longer response time [22.2±0.4 ms for (3,3), 10.0±0.3 ms for (1,3), and 10.3±0.2 ms for (3,4), respectively], supporting the ionic doping/dedoping pathway start from the channel edge (Fig. 3e and Supplementary Table 2).

While increasing switching frequency from 11.1 Hz to 111.1 Hz, the overall baseline shifts are enhanced, where (1, 3) shows an extraordinary shift of 80.2 nm, while only 46.5 nm and 31.3 nm at (3,4) and (3, 3), respectively, are observed (Fig. 3f,g). In fact, tremendously pronounced swellings at top electrode edge without bottom electrode underneath [(1,y) and (6,y), y = 1:4] are consistently observed, which are 184.2±26.1 nm, 183.4±24.2 nm, and 107.5±31.1 nm for switching frequencies of 3.3 Hz, 11.1 Hz and 111.1 Hz, respectively, while corresponding averaged swelling magnitudes in the channel are only 125.1±22.5 nm, 118.8±21.1 nm, and 33.5±20.1 nm (Fig. 3c, Extended Data Fig.4a,b, and Supplementary Videos 1 to 3). This phenomenon directly indicates that excessive water molecules in the electrolyte solution have been injected into the channel edge (especially the part without the bottom electrodes) during vOECT operation, which could impair cycling stability. In fact, similar swell mapping characteristics are observed in all other polymer channels (Supplementary Videos 4 to 12). Specifically, despite varying swelling magnitudes across different materials, they all exhibit pronounced swelling at the top-electrode edge (Extended Data Fig.5). Especially, up to 380 nm total swelling is observed in p(g2T-T):GDA with an original film thickness of only 80 nm.

To probe the origin of such huge swelling at the top electrode edge, source and drain voltages (V_S and V_D) are both set to 0 V during V_G switching. As shown in Extended Data Fig.6, for a representative point (3,1), which is located at the top electrode edge with the bottom electrode underneath, enhanced swelling magnitudes are always observed for different channels when no source/drain bias is applied. Especially, apparent baseline shifts are observed with p(g2T-T) and gDPP-g2T-based channels. However, as long as V_D is biased, no baseline shifts and reduced swelling magnitudes are observed. It is suggested that, when a vertical electric field is applied, i.e., a weak Coulombic force is formed between two electrodes, channel swelling can be effectively suppressed. Additionally, after 1000 switching cycles in the absence of the Coulombic force in a BBL-based vOECT, distinct wrinkles (> 400 nm in height) in the channel region can be observed by optical microscopy and AFM (Supplementary Figs. 8).

Modulation of the swelling in vOECT towards high-fidelity OECN

Consequently, to suppress swelling at the critical top electrode edge region, the encapsulation pattern is re-designed by precisely aligning the opening width with that of the bottom electrode(Fig. 4a and Supplementary Fig. 9). As shown in Supplementary Figs. 10 and 11, swelling is effectively suppressed at

the top electrode edge, leading to an overall 21% reduction in swelling amplitude compared to that without optimized encapsulation. Consequently, cycling stabilities of vOECTs with different channel materials can be further enhanced, along with a minor effect on transistor transient characteristics (Fig. 4. b,c, and Extended Data Fig.7). Note, the detailed cycling process can be found in Methods. For instance, with switching V_G from 0.4 V to -0.6 V, 99.7% I_D retention after 100k cycles (1.4 hrs) is obtained for the optimized p(g2T-T):DA-based vOECT, markedly surpassing 63.84% I_D retention in the conventionally encapsulated counterpart. Note, channels with GDA crosslinker typically exhibit lower stabilities but faster transient responses than those with DA, which is attributed to the hydrophilic ethylene glycol chain in GDA that facilitates swelling. Moreover, extensively continuous cycling characteristics on vOECTs based on p(g2T-T):DA, Homo-gDPP:DA, and BBL are conducted (Fig. 4d and Extended Data Fig.8). These devices demonstrate stable full switching cycles for 15 million (208.3 hours) and 2.8 million (62.2 hours) for p-type and n-type OECTs, respectively. Note that, in these cycling stability measurements, no measures (for instance, degassing) to avoid side reactions are adopted[31], while these stabilities are still among the highest reported for OECTs (Fig. 4e and Supplementary Table 3) [28,31-41].

Next, such vOECTs are utilized for the construction of a vertically stacked organic electrochemical neuron (vOECN) based on the axon hillock (A-H) circuit. First, vertically stacked complementary inverters are fabricated using gDPP-g2T:DA and Homo-gDPP:DA as p-/n-type channels, respectively (Supplementary Figs. 12). Corresponding transfer and transient curves demonstrate on/off current ratios of ~10^6 and transient times of ~ 1 ms, which enable inverters with robust logic switching up to 1250 Hz. Then, vOECNs are further integrated (Fig. 4f, Supplementary Figs. 13, and detailed fabrication process can be found in Methods) and characterized. The present leaky integrate-and-fire (LIF) OECN structure mimics the action potential spikes generated by nerve cells[42], making the dynamic range of spike frequencies profoundly influence the diversity of neural behaviors[40,43-45]. Thus, spike frequency coverage of the developed vOECNs is systematically investigated.

The pulse frequency (f_{fire}) of vOECNs can be regulated by the charge-discharge rate of membrane capacitance in the circuit, a process that V_{DD}, C_{mem}, and I_{IN} can all manipulate[2,40,46]. As shown in Fig. 4g, under fixed conditions of C_{mem} = C_f = 500 pF and I_{IN} = 1 μA, f_{fire} decreases monotonically from 92.6 Hz to 33.6 Hz as V_{DD} rises from 0.6 V to 0.8 V. This behavior originates from the lower V_{DD} reducing the threshold voltage, enabling capacitors to charge to the threshold more rapidly and trigger spikes. I_{IN} also significantly influences the charge-discharge dynamics of C_{mem}, where higher I_{IN} accelerates the process (Fig. 4h). Furthermore, by adjusting C_{mem}, the frequency range of vOECNs extends from 0.5 Hz to 1.1 kHz, representing the widest bandwidth reported for OECT-based OECNs to date, fully encompassing the firing spectrum of human neurons (Fig. 4i, Extended Data Fig.9, and Supplementary Table 4)[2,40,42,46-52]. This is attributed to high-resolution channel packaging (55 × 105 μm) that reduces parasitic capacitance and integration of smaller external capacitors.

Lastly, the stability of vOECN is characterized. Fig. 4j demonstrates that vOECN maintains 100% fire voltage (V_{fire}) retention after 155,000 switching cycles, an excellent performance attributed to the optimized encapsulation. Furthermore, no performance degradation (100% V_{fire} and f_{fire} retention) is observed after the vOECN is immersed in 1× PBS solution for 15 days, confirming sustained operation throughout the testing period (Fig. 4k). The combined wide-range spiking capability and exceptional stability metrics fully demonstrate the significant application potential of vOECN in the fields of neuromorphic electronics and brain-computer interfaces.

This study introduces a novel in-operando channel swelling monitoring tool, LDV-M, which is currently the sole platform that enables real-time dynamic swelling monitoring in operando conditions. High spatiotemporal resolution swell mapping can be easily accessed, enabling direct observation of swelling behaviors in OMIEC-based vOECTs under diverse conditions (different channel materials, switching frequencies, driving voltages, cycling states, etc.). Apparent differences in swelling magnitudes (from several nm to hundreds of nm) in different channel materials are observed, along with clear ionic doping paths from the vertical channel edge to the channel center. Especially, significant lateral top electrode edge swelling is identified as the primary cause of device degradation, while Coulombic forces between vertically stacked source/drain electrodes effectively suppress such swelling. Consequently, a precise device encapsulation configuration is implemented, enhancing vOECT cyclic stability beyond 15 million full cycles (the highest reported to date). Furthermore, vOECN based on such architecture achieves the broadest f_{fire} range from 0.5 Hz to 1.1 kHz, along with 100% spike sustaining after 155,000 cycles and 15-day immersion in 1×PBS. Overall, LDV-M provides a novel in situ characterization methodology for dissecting OECT operational mechanisms and offers a practical research tool for OMIEC-based bionic neural engineering.

Methods

Materials

gDPP-g2T, Homo-gDPP, DA, and GDA were synthesized via previously reported methods[25,27,28]. p(g2T-T) was purchased from Derthon Optoelectronics Materials Science Technology Co., Ltd. BBL, Ag/AgCl paste, Isopropyl alcohol, methanesulfonic acid, and chloroform were purchased from Sigma-Aldrich. Phosphate buffer saline (0.01 M PBS) supplied by Solarbio was used as the electrolyte.

Semiconductor solution preparation

gDPP-g2T, Homo-gDPP, DA, and GDA were first dissolved in chloroform, respectively, with a concentration of 20 mg ml^{-1}. p(g2T-T) was dissolved in chloroform at a concentration of 10 mg ml^{-1}. Then those solutions were filtered through a 0.45 μm polyvinylidene difluoride filter. Subsequently, gDPP-g2T, Homo-gDPP, or p(g2T-T) solution was mixed with DA or GDA solution in a mass ratio of 5:1. For BBL, it was dissolved in methanesulfonic acid at a concentration of 15 mg ml^{-1}.

Vertical OECT fabrication

The vOECT fabrication process is illustrated in Supplementary Figs. 2. First, 3 nm of Cr and 130 nm of Au (rate approximately 0.5–2.0 Å s^{-1}) were thermally evaporated on a precleaned Si/100 nm SiO_2 wafer with a shadow mask as the bottom source electrode and gate electrode. Then, the substrates with the bottom electrodes underwent a 15-minute UV-ozone cleaning process. Subsequently, solutions of BBL, Homo-gDPP:GDA, Homo-gDPP:DA, p(g2T-T):GDA, p(g2T-T):DA, gDPP-g2T:GDA, and gDPP-g2T:DA, respectively, were spin-coated at 2,500 rpm for 30 seconds. The films with crosslinkers were then cross-linked by exposure to 365 nm UV light (125 mW/cm^2) for 2 min; while BBL film was first subjected to a 15-minute immersion in deionized water and subsequently blow-dried with a nitrogen gun. Then, the semiconductor layer was patterned using an LPKF protolaser R4 at 500 kHz with a power of 0.13 W. Then, the top drain electrode (130 nm Au) was thermally evaporated (rate approximately 0.5–2.0 Å s^{-1}) with a shadow mask while maintaining the substrate at a temperature of ~ 5 ℃ with a back water-cooling system. Subsequently, an SU8-2002 solution was spin-coated onto the substrate at 2,000 rpm for 60 s. The sample was then heated on a 95 °C hotplate for 80 s, exposed to light (28 mW/cm²) for 20 s using a shadow mask, and reheated on the 95 °C hotplate for 80 s. The encapsulation layer was then patterned by developing the SU8 in a developer for 10 s, rinsing in isopropanol for 10 s, and blow-drying. Finally, an Ag/AgCl paste was applied onto the gate electrode and vacuum-dried for 30 min, the device was semi-sealed with glass, and 10–30 µL of 1× PBS solution was introduced into the semi-sealed region.

Planar OECT fabrication

First, 3 nm of Cr and 50 nm of Au (rate approximately 0.5–2.0 Å s^{-1}) were thermally evaporated on a precleaned Si/100 nm SiO_2 wafer with a shadow mask as the source, drain, and gate electrodes. Then, the substrates with the electrodes underwent a 15-minute UV-ozone cleaning process. Subsequently, solutions of BBL and p(g2T-T):GDA were spin-coated at 2,500 rpm for 30 seconds. The films with crosslinkers were then cross-linked by exposure to 365 nm UV light (125 mW/cm^2) for 2 min; while BBL film was first subjected to a 15-minute immersion in deionized water and subsequently blow-dried with a nitrogen gun. Then, the semiconductor layer was patterned using an LPKF protolaser R4 at 500 kHz with a power of 0.13 W. Subsequently, an SU8-2002 solution was spin-coated onto the substrate at 2,000 rpm for 60 s. The sample was then heated on a 95 °C hotplate for 80 s, exposed to light (28 mW/cm²) for 20 s using a shadow mask, and reheated on the 95 °C hotplate for 80 s. The encapsulation layer was then patterned by developing the SU8 in a developer for 10 s, rinsing in isopropanol for 10 s, and blow-drying. Finally, an Ag/AgCl paste was applied onto the gate electrode and vacuum-dried for 30 min, the device was semi-sealed with glass, and 10–30 µL of 1× PBS solution was introduced into the semi-sealed region.

Vertical OECN fabrication

vOECN fabrication process can be found in Supplementary Fig. 13. First, 3 nm of Cr and 80 nm of Au (rate approximately 0.5–2.0 Å s^{-1}) were thermally evaporated on a precleaned glass substrate with a shadow mask as the bottom electrode. gDPP-g2T:DA blend solution was spin-coated at 2,500 rpm for 20 s. The film was then cross-linked by exposure to 365 nm UV light (125 mW/cm^2) for 2 min, and the semiconductor layer was patterned using an LPKF protolaser R4 at 500 kHz with a power of 0.13 W. Next, the middle drain electrode (80 nm Au) was then thermally evaporated (rate approximately 0.5–2.0 Å s^{-1}) with a shadow mask while maintaining the substrate at a temperature of ~ 5 °C with a back water-cooling system. Then, Homo-gDPP:DA blend solution was spin-coated and photopatterned using the same protocol as the p-type material to form the n-type channels in the amplifier block and T_{reset} regions. The top electrode (GND) was fabricated by thermal evaporation of an 80 nm Au layer through a shadow mask. To expose active channel areas, gate electrodes, and capacitor connection points, the substrate was coated with an SU8-2002 solution via spin-coating at 2,000 rpm for 60 s, followed by a 95 °C hotplate bake for 80 s. The SU8 layer was then exposed to 28 mW/cm² UV light through a shadow mask for 20 s and subjected to a post-exposure bake at 95 °C for 80 s. Patterning development was achieved by immersing the sample in SU8 developer for 10 s, rinsing with isopropanol for 10 s, and drying with compressed air to form the encapsulated structure. Gate electrodes of the amplifier block and T_{reset} were coated with Ag/AgCl paste and vacuum-dried for 30 min. Capacitors were then attached using conductive silver paint. Finally, 1× PBS electrolyte was dropped onto the gate electrodes and adjacent active channel regions to complete device fabrication.

LDV-M platform configuration

The LDV-M employed a 633 nm wavelength Doppler laser vibrometer (Polytec Inc., VibroFlex Compact), with a ×20 objective lens (VIB-A-20×) mounted on the laser probe to focus the laser spot to be 3 μm diameter. vOECT electrodes were clamped by a 3M chip testing fixture (which is further connected with a semiconductor analyzer) and fixed onto a custom-designed planar electrical positioning stage with micron-level positioning accuracy and multi-axis adjustability (x, y, z, pitch angle). The laser is incident perpendicularly onto the sample surface, while the laser spot position on the surface is real-time monitored by a CMOS color camera integrated into the vibrometer. Additionally, a semiconductor parameter analyzer (FS-Pro) drives the OECT and acquires its electrical signals simultaneously, while the laser vibrometer collects swelling signals. Notably, both the vibrometer and the planar electrical positioning stage are mounted on the same optical table to enhance the signal-to-noise ratio.

Electrical measurements

Under ambient conditions, the electrical properties of vOECTs, inverters, and vOECNs were characterized using an FS-Pro (PDA) semiconductor analyzer and a probe station (Chengdu Chiptest Technology Co., LTD.).

For cycling stability test: first, a transfer curve of the device was measured; then, a rectangular wave pulse with fixed frequency and amplitude was applied to the V_G for cyclic stability testing, while V_D was

0.1 V and -0.1 V for n-type and p-type OECTs, respectively. This process was repeated multiple times to enable larger cycling numbers and time. Specifically, for p(g2T-T):DA and p(g2T-T):GDA-based vOECT, V_G = 0.4 to -0.6 V, frequency = 20 Hz; for BBL-based vOECT, V_G = -0.3 to 0.5 V, frequency = 12.5 Hz; for Homo-gDPP:DA-based vOECT, the test conditions in Fig. 4c,d were V_G = 0 to 0.9 V at 25 Hz and V_G = 0 to 0.8 V at 20 Hz, respectively; for Homo-gDPP:GDA-based vOECT, V_G = 0 to 0.9 V, frequency = 25 Hz; for gDPP-g2T:DA and gDPP-g2T:GDA-based vOECT, V_G = -0.3 to -0.6 V, frequency = 25 Hz.

In Fig. 4c and Extended Data Fig. 7, transfer curves were gathered every 20,000 switching cycles for vOECTs using p(g2T-T):DA/GDA, Homo-gDPP:DA/GDA, or gDPP-g2T:DA/GDA, and every 5,000 switching cycles for BBL-based vOECT. In Fig. 4d and Extended Data Fig. 8, transfer curves were collected every 100,000 switching cycles for vOECTs based on p(g2T-T):DA, and every 50,000 switching cycles for those based on Homo-gDPP:DA and BBL. Notably, to maintain a relatively stable 1× PBS electrolyte concentration, a PDMS well was placed on top of the active region of the device, and ~50 μL of 1× PBS solution was added to restrict electrolyte displacement and slow down electrolyte evaporation.

For the voltage transfer characteristic measurement of inverters, the V_{IN} was supplied by the FS-Pro with a step interval of 2 mV, and the output voltage was also recorded by the FS-Pro (PDA). To characterize the frequency response of the inverters, the FS-Pro was used to apply V_{IN} voltages with a fixed amplitude range of 0 to 0.7 V and varying frequencies (50 Hz to 1.25 kHz), while the output characteristics were recorded.

For the electrical characterization of vOECNs, the FS-Pro provided different constant I_{IN} = 0.05 to 50 μA and V_{DD} = 0.6 to 0.8 V, with a sampling frequency of 50,000 samples per second during testing. To evaluate the cyclic stability of vOECNs, a constant V_{DD} = 0.7 V and I_{IN} = 1.1 μA were applied via the FS-Pro, and the output characteristics were recorded. Notably, three independent PDMS wells were designed at the electrolyte droplet positions to ensure an adequate volume of 1× PBS solution and prevent short circuits caused by electrolyte mixing. For the storage stability test of vOECNs, a 3D-printed sealed container was employed to house the vOECN, which was immersed in 1× PBS within the container. Before each test, the vOECN was removed, rinsed with deionized water, dried using a nitrogen gun, and then characterized.

Vertical OECT swelling measurements

For swelling testing of vOECTs, the laser spot position on the vOECT device surface was first observed and adjusted using a CMOS color camera to determine the measured point position (Polytec Inc., VibroFlex Compact). Subsequently, the vOECT is driven by the Fs-Pro semiconductor analyzer, and V_G supplied by Fs-Pro is also synchronized with the laser vibrometer. Note, V_G switching frequencies of 3.3 Hz, 11.1 Hz, and 111.1 Hz are chosen to minimize the interference of the power line interference (50 Hz).

For swell mapping of vOECT channels, the laser spot was first positioned at the lower-left corner of the OECT top electrode (1,1) using a CMOS color camera. A custom-designed planar electrical positioning

stage controlled by LabView was used to perform a 6×4 scanning matrix with a 16 μm step size: 5 steps right (80 μm), return to the origin, 1 step up, and 3 steps right (50 μm), scanning from the lower-left to upper-right of the electrode. During scanning, Fs-Pro generated rectangular wave V_G signals to switch the device on/off state and record I_D; the laser vibrometer collects the corresponding swelling and V_G signals, while the CMOS camera records the position of the laser spot at each step.

Planar OECT swelling measurements

For swelling testing of planar OECTs, the laser spot position on the planar OECT device surface was first observed and adjusted using a CMOS color camera to determine the measured point position (Polytec Inc., VibroFlex Compact). Subsequently, the planar OECT is driven by the Fs-Pro semiconductor analyzer. Note, V_G switching frequency is 3.3 Hz.

OECT swelling data analysis

For the collected vOECT/planar OECT swelling data, the acquired signals were filtered using a band-stop filter at 24-26 and 49-51 Hz to minimize power line interference (50 Hz). Additionally, as both the source meter (Fs-Pro) and laser vibrometer record V_G voltage signals, synchronization of the vOECT switching current with the swelling signal was achieved by aligning the V_G voltage traces from both devices.

Swelling magnitudes of the vOECT channels contain two parts: active swelling and baseline shift (Fig. 2c and Fig. S6). When analyzing the degree of doping and dedoping of films, swelling from baseline shift is excluded to focus solely on active swelling signals. All swelling data analyses, except those labeled "active swelling", include both baseline shift and active swelling (e.g., 3D swelling distribution maps of devices and swelling across multiple device cycles).

EQCM measurements

EQCM measurements were performed using a gold-coated quartz crystal sensor and a quartz crystal microbalance (QCM922A, Princeton Applied Research). The polymer film was patterned to define an active gold area (~0.2 cm²). Before film deposition, the crystal's fundamental frequency was tested in air and electrolyte (0.01 M PBS) solution environments. After spin-coating and patterning the semiconductor film onto the crystal, the fundamental frequency was retested in the same environments to calculate passive swelling. Finally, the crystal was connected to an Fs-pro semiconductor parameter analyzer via a chip testing fixture, with the Fs-pro generating rectangular wave voltage signals to induce film doping/dedoping while recording current signals. Previous studies have shown that ~100 nm-thick polymer films can be considered "rigid" films[18], so in this work, mass changes were calculated using the Sauerbrey equation:

$$\frac{\Delta m}{A} = -\Delta f_n \frac{\rho_q V_q}{2 f_0^2 n} \approx \frac{-\Delta f_n}{n} 17.9 \text{ ng cm}^{-2}$$

where Δf_n is the frequency shift of the nth overtone, A is the sensor active area, ρ_q is the density of quartz, Vq is the shear wave velocity in quartz, f_0 is the fundamental frequency and n is the overtone number (fixed at 1). Relative mass change was calculated as the percentage change in total mass relative to the dry mass:

$$Relative\ mass\ change = \frac{m_{total} - m_{dry}}{m_{dry}} \times 100\ \%$$

AFM Measurements

The film thickness was measured using atomic force microscopy (AFM) in tapping mode, with a Park NX7 AFM equipped with a silicon tip. Specimens for film thickness measurement were prepared following the same fabrication process as that used for the semiconductor layers in each vOECT. Then, the film surface was scratched with tweezers. Subsequently, AFM was used to measure the height difference between the two sides of the scratch, which was taken as the thickness of the film.

Declarations

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Author contributions: Z.D., C.C., and W.H. conceived the idea and designed the experiments. Z.D. and S.Z. performed OECT fabrication and measurement. Z.D., J.W., M.X., L.H., Y.C., C.C., and W.H. constructed the swelling system. Z.D., S.Z., and D.Z. performed the OECT swelling measurement. Z.D. performed OECN fabrication and measurement. Z.D. and D.L. produced the swelling animation. Y.Z. performed the AFM measurement. Z.D., S.Z., J.Z., and L.-W.F. performed the EQCM

measurement. J.C. synthesized the Homo-gDPP and gDPP-g2T. Y.L. and L.-W.F. synthesized the DA and GDA. Z.D. and W.H. wrote the manuscript. All the authors reviewed and approved the manuscript.

Competing interests: Z.D., C.C., and W.H. are inventors on patent applications related to the LDV-M platform and designs of high-fidelity OECTs.

Data Availability: The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Figures

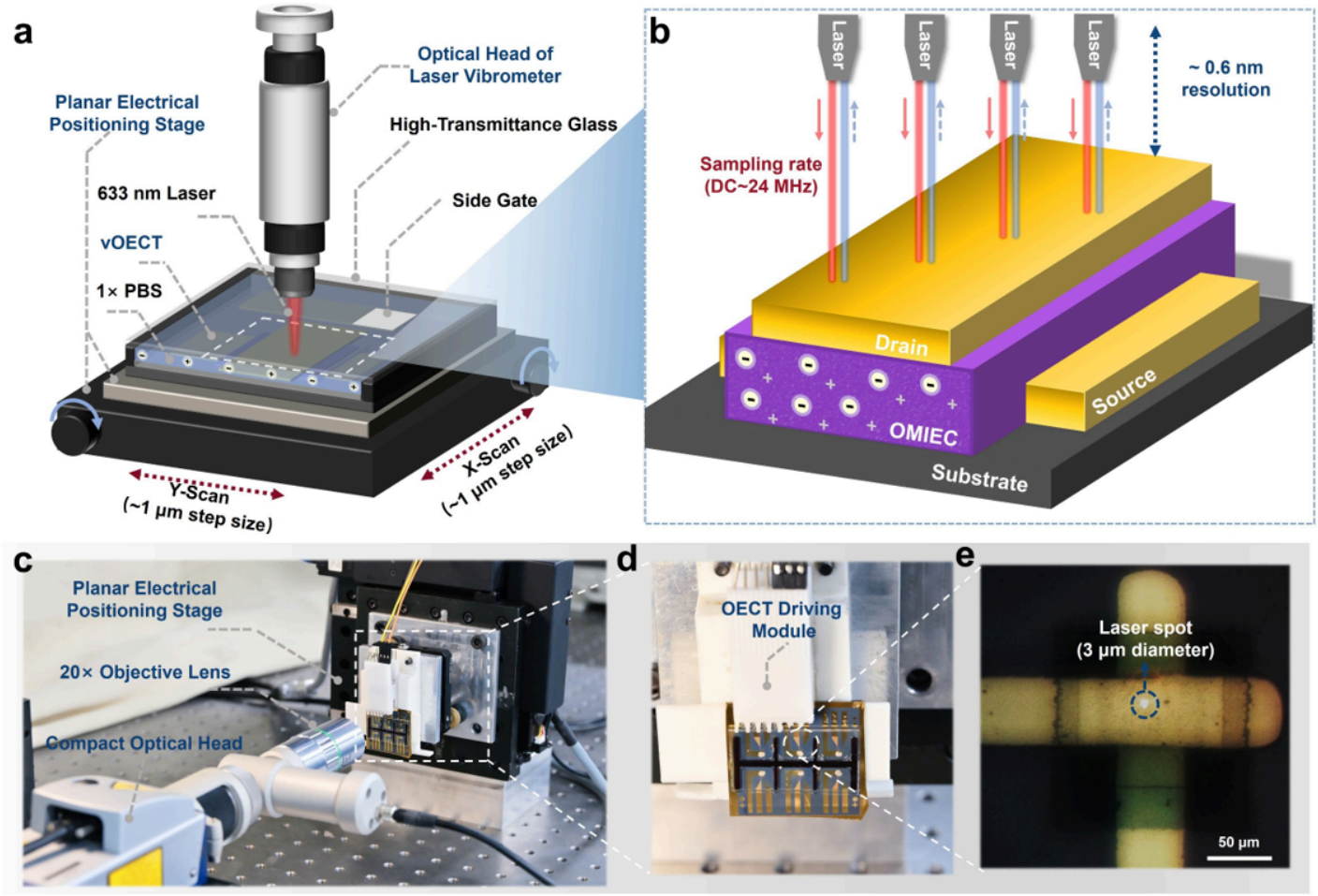


Figure 1

System configuration of the laser Doppler vibrometer with mapping (LDV-M). a, Main components of the LDV-M. **b,** Schematic diagram of LDV linked with planar electrical positioning stage for channel swelling monitoring. **c,** Optical image of the LDV-M, along with the zoomed-in **(d)** OECT mounting module and **(e)** transistor channel with a measuring laser spot focused on.

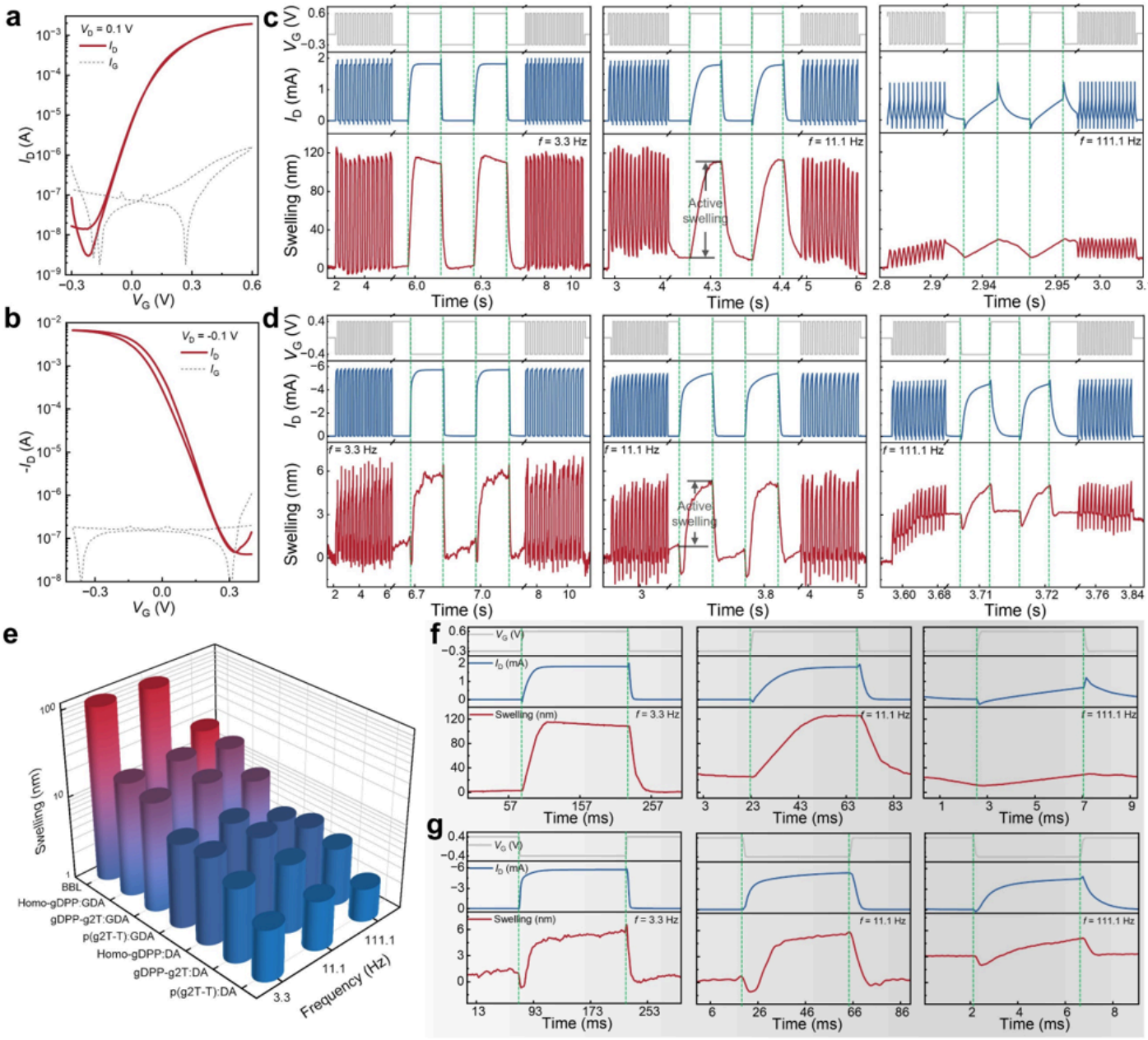


Figure 2

Single-point swelling monitoring. Transfer curves of vOECTs based on **(a)** BBL or **(b)** p(g2T-T):DA. Single point swelling behaviors of vOECT channels based on **(c)** BBL or **(d)** p(g2T-T):DA under different switching frequencies. **e,** Active swelling magnitudes of the vOECT channel center based on different channels under varying frequencies. Representative swelling characteristics within a single switching cycle under different frequencies for vOECTs based on **(f)** BBL or **(g)** p(g2T-T):DA. V_D = 0.1 V (n-type) or -0.1 V (p-type).

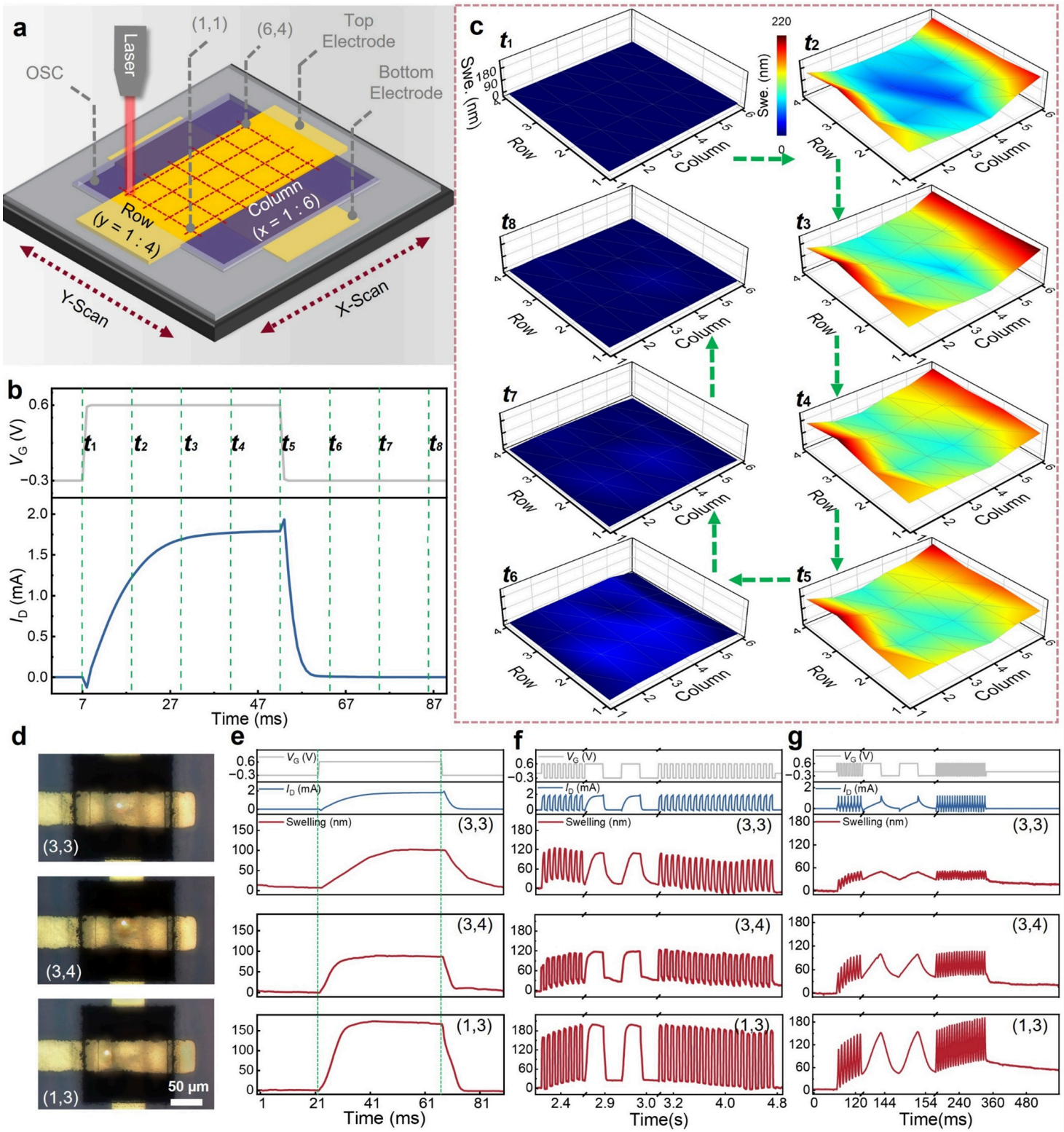


Figure 3

Swell mapping of vOECT channel. a, Schematic of laser light spot distribution. **b,** An I_D switching cycle along with **(c)** the corresponding swell mapping frames of a BBL-based vOECT channel. **d,** Representative mapping point locations, along with **(e)** the swelling behavior during a single switching cycle and **(f,g)** multiple swelling cycles. Switching Frequencies are 11.1 Hz and 111.1 Hz for **(e,f)** and **(g)**, respectively.

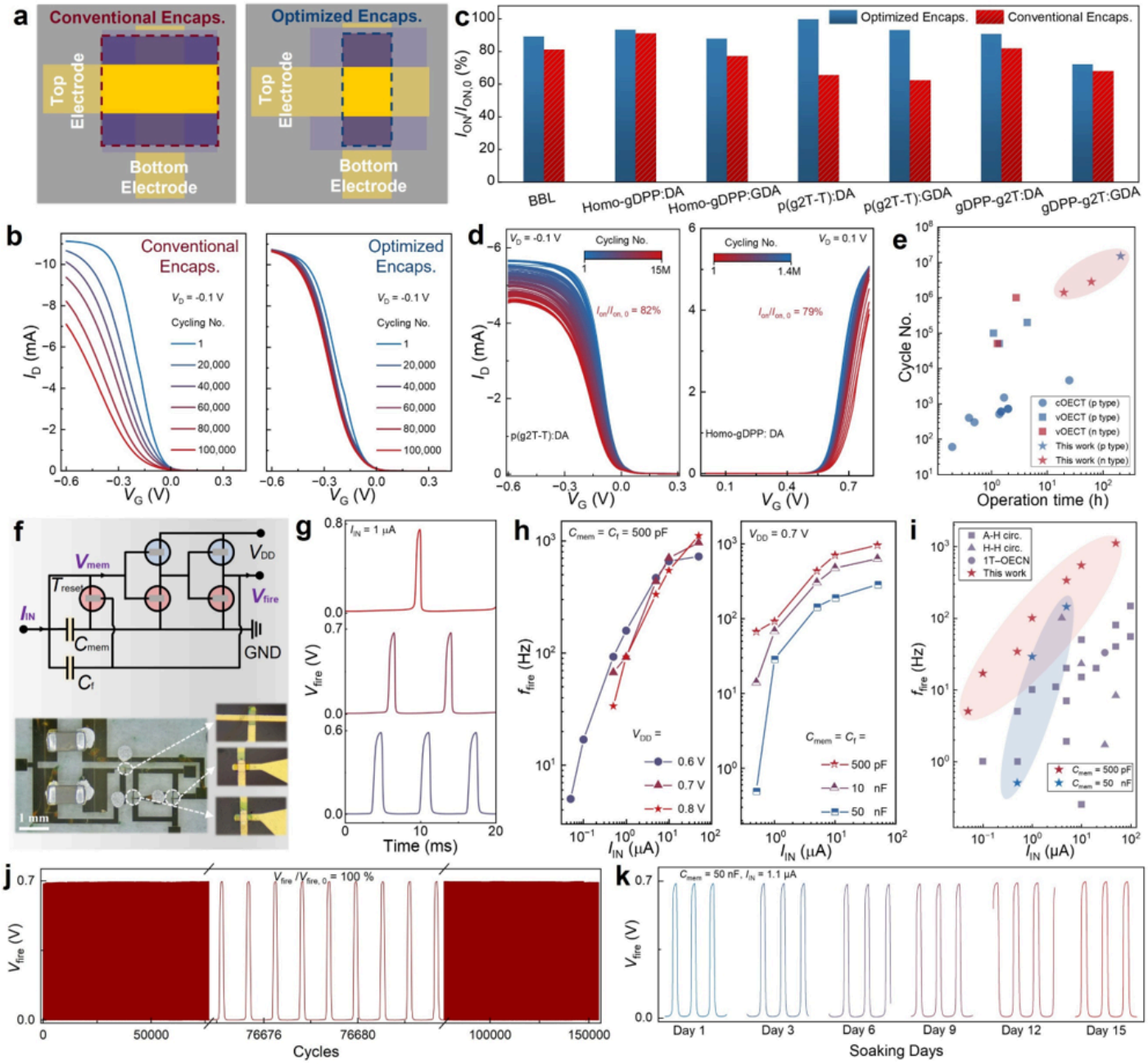


Figure 4

Design and characteristics of vOECTs and high-fidelity vOECNs. a, Top view of the channel with conventional or optimized encapsulation. **b,** Transfer curves of vOECTs based on p(g2T-T):DA with two different encapsulations after indicated cycle stability measurements. **c,** On current retention ratio for vOECTs based on different channels with different encapsulation methods after 100,000 switching cycles. **d,** Transfer curves of vOECTs based on p(g2T-T):DA and Homo-gDPP:DA after indicated switching cycles. **e,** Comparison of cycling numbers and operation times of reported OECTs. **f,** Circuit diagram and optical image of a vOECN. **g,** vOECN fire frequency modulation under three different V_{DD} configurations and **(h)** the coverage ranges of f_{fire} at different I_{IN} with varying V_{DD} (V_{DD} =0.6 V, 0.7 V, and 0.8 V; C_{mem} = C_f = 500 pF) or with varying capacitance (C_{mem} = C_f = 500 pF, 10 nF, or 50 nF; V_{DD} =0.7 V). **(i)** Fire frequency

versus I_{IN} in this work compared with those from reported OECNs. **(j)** Cycling stability and **(k) s**torage stability of the vOECN with a switching frequency of 34 Hz (V_{DD} = 0.7 V, I_{IN} = 1.1 μA, and C_{mem} = 50 nF).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- MovieS11gDPPg2T11.1hz.mp4
- MovieS7pg2TT3.3hz.mp4
- MovieS10gDPPg2T3.3Hz.mp4
- MovieS2XXXBBL11.1HzXXX.mp4
- MovieS8pg2TT11.1hz.mp4
- MovieS1XXXBBL3.3HzXXX.mp4
- MovieS9pg2TT111.1hz.mp4
- MovieS12gDPPg2T111.1hz.mp4
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- MovieS3XXXBBL111.1HzXXX.mp4
- MovieS5XXXHomogDPP11.1hzXXX.mp4
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